RETAIL SALES FORECASTING USING LSTM

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Declaration

The Project Report entitled **"Retail sales forecasting using LSTM"** is a record of Bonafide work of team members **Chundru Harshitha – 2320030231, Ramagiri Akshaya – 2320030224, Mamindla Srihitha - 2320030248, Lam Divya Sri – 2320030250** submitted in partial fulfilment for the award of B. Tech in Computer Engineering to K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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Certificate

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ABSTRACT

Retail sales forecasting is a critical task for effective inventory management and business decision-making. This project presents a data-driven approach to predict future sales using a Long Short-Term Memory (LSTM) based time series model. The system takes historical retail data as input and processes it through stages including data preprocessing, normalization, and sequence generation to capture temporal patterns.

An LSTM neural network is employed to learn long-term dependencies in sales data and generate accurate predictions. The model forecasts future sales for the next 7, 15, or 30 days based on past trends. The system also supports both overall and category-wise forecasting by allowing users to filter data dynamically, enabling more granular analysis.

A user-friendly interactive dashboard is developed using Streamlit, where users can upload datasets, visualize actual versus predicted sales, and view future forecasts through graphical representations. Additionally, the system provides intelligent insights such as trend analysis, peak sales periods, and automated inventory recommendations (increase, reduce, or maintain stock) based on predicted demand.

This project demonstrates how deep learning techniques can be effectively applied to real-world retail scenarios, helping businesses optimize inventory, reduce losses, and improve operational efficiency.

INTRODUCTION

In today's competitive retail environment, accurate sales forecasting plays a crucial role in effective inventory management and strategic decision-making. Businesses must anticipate customer demand to avoid issues such as overstocking, stock shortages, and financial losses. Traditional forecasting methods often fail to capture complex patterns and seasonality present in real-world data, making advanced techniques necessary.

With the rapid growth of data availability and machine learning technologies, time series forecasting using deep learning has gained significant importance. Among these techniques, Long Short-Term Memory (LSTM) networks, a type of Recurrent Neural Network (RNN), are particularly effective in handling sequential data and capturing long-term dependencies. LSTM models are well-suited for retail sales prediction, as they can learn patterns from historical sales data and generate accurate future forecasts.

This project focuses on developing a retail sales forecasting system using an LSTM-based model. The system processes historical sales data through preprocessing steps such as data cleaning, normalization, and sequence generation. The trained model predicts future sales trends and enables users to visualize actual and predicted values through an interactive dashboard built using Streamlit. Additionally, the system supports both overall and category-wise forecasting, allowing users to analyze specific product categories. It also provides insights such as trend analysis, peak sales periods, and inventory recommendations to assist businesses in making informed decisions.

Overall, this project demonstrates the practical application of deep learning in retail analytics, helping improve forecasting accuracy, optimize inventory management, and enhance operational efficiency.

LITERATURE SURVEY

The rapid growth of retail businesses and the increasing availability of transactional data have led to a significant rise in the volume of sales data generated daily. Traditional forecasting methods are often insufficient to handle such large-scale and dynamic datasets. As a result, advanced data analytics and machine learning techniques have emerged as essential tools for predicting sales trends and supporting business decisions. Several researchers have explored the use of time series analysis and deep learning models to improve forecasting accuracy in retail systems.

Retail sales forecasting has become an important research area due to its role in inventory management, demand planning, and operational efficiency. Accurate prediction of sales helps businesses avoid overstocking and stock shortages while improving customer satisfaction. With the advancement of data-driven technologies, organizations are increasingly relying on intelligent forecasting models to analyze historical data and predict future demand.

Time Series Forecasting in Retail Systems

Time series forecasting techniques have been widely used in retail analytics. Traditional methods such as Moving Averages, ARIMA, and Exponential Smoothing have been applied to predict sales trends. These models are effective for linear data but often fail to capture complex patterns, seasonality, and sudden fluctuations present in real-world retail data. Researchers have found that these limitations reduce the accuracy of predictions, especially when dealing with large and highly dynamic datasets.

Deep Learning Approaches for Sales Forecasting

With the limitations of traditional models, deep learning techniques have gained popularity in recent years. Long Short-Term Memory (LSTM), a type of Recurrent Neural Network (RNN), has proven to be highly effective in handling sequential and time-dependent data. LSTM networks are capable of learning long-term dependencies and capturing non-linear relationships in data, making them suitable for retail sales forecasting.

Several studies have shown that LSTM-based models outperform traditional statistical approaches in predicting sales trends. These models are able to adapt to changing patterns and provide more accurate forecasts, especially when dealing with seasonal variations and complex customer behavior.

Category-Based and Multi-Dimensional Forecasting

Recent research has also focused on category-based forecasting, where sales are analyzed at a more granular level, such as product categories or store locations. This approach allows businesses to gain deeper insights into specific segments and make more targeted decisions. By filtering and analyzing data based on categories, forecasting models can provide more precise and actionable results.

Researchers have demonstrated that combining category-level analysis with time series forecasting improves the overall effectiveness of retail analytics systems. It enables better inventory planning and helps identify high-performing product categories.

Visualization and Decision Support Systems

Data visualization plays a crucial role in interpreting forecasting results. Visual tools such as line graphs, trend charts, and dashboards help in understanding sales patterns and comparing actual versus predicted values. Interactive dashboards allow users to explore data dynamically and make informed decisions.

Modern tools like Streamlit enable the development of user-friendly dashboards that integrate machine learning models with visualization. These systems enhance decision-making by providing real-time insights into sales trends, peak seasons, and future forecasts.

Inventory Optimization and Decision Making

An important application of sales forecasting is inventory optimization. By analyzing predicted sales trends, businesses can make decisions regarding stock levels. If future demand is expected to decrease, stock levels can be reduced to avoid losses. Conversely, increasing demand indicates the need to maintain or increase inventory levels.

Several studies highlight that integrating forecasting models with decision support systems improves operational efficiency and reduces costs. Automated insights such as trend detection and peak season identification further assist businesses in strategic planning.

Conclusion

The literature clearly indicates a shift from traditional statistical methods to advanced deep learning approaches for retail sales forecasting. LSTM models provide better accuracy, improved handling of complex patterns, and enhanced adaptability to dynamic data. Additionally, the integration of category-based analysis and interactive visualization tools has significantly improved the effectiveness of forecasting systems.

This project builds upon these research findings by implementing an LSTM-based retail sales forecasting system with an interactive dashboard. By supporting both overall and category-wise

analysis, generating future predictions, and providing inventory recommendations, the system demonstrates the practical application of modern deep learning techniques in retail analytics.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Specifications:

The Retail Sales Forecasting using LSTM-Based Time Series Data Pipeline project does not require high-end hardware, as the computations are moderate and can be executed on a standard system. The system is primarily used for data preprocessing, model training, and visualization through a web-based interface.

Processor

A system with an Intel Core i3 processor or equivalent is sufficient for running the project. However, for better performance and faster model training, an Intel Core i5 or higher is recommended. Since LSTM models involve iterative computations, a better processor helps in reducing training time.

RAM:

A minimum of 4 GB RAM is required to run the application, including the Streamlit interface and Python environment. However, 8 GB RAM or above is recommended for smoother performance, especially when handling larger datasets and training deep learning models.

Storage:

The project requires minimal local storage. At least 500 MB of free disk space is sufficient for:

- Python installation
- Required libraries (TensorFlow, Pandas, etc.)
- Temporary files and datasets

If larger datasets are used, additional storage may be required.

Graphics:

No dedicated graphics card is required. Integrated graphics available in standard processors are sufficient, as the project focuses on data analysis and visualization rather than high-end graphical processing. However, systems with GPU support can accelerate deep learning model training.

Operating System:

The project is compatible with:

- Windows
- Linux
- macOS

Since the application is developed using Python and Streamlit, it can run on any operating system that supports these technologies.

Software Requirements

The project uses open-source tools and libraries for data processing, machine learning, and visualization.

Programming Language:

- Python (Version 3.8 – 3.10 recommended)

Python is used for implementing data preprocessing, LSTM model development, and building interactive dashboards.

Libraries and Frameworks:

The following Python libraries are used:

- TensorFlow – For building and training the LSTM model
- Pandas – For data manipulation and preprocessing
- NumPy – For numerical computations
- Matplotlib – For plotting graphs and visualizing results
- Scikit-learn – For data scaling (MinMaxScaler)

These libraries are essential for implementing the complete time series forecasting pipeline.

Development Environment:

- Visual Studio Code (VS Code) – For writing and managing code
- Jupyter Notebook / Google Colab (Optional) – For experimentation and testing
- Command Prompt / Terminal – For running Streamlit applications

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Web Application Framework:

- Streamlit is used to develop an interactive dashboard that allows users to:
- Upload datasets
- Select features (date, sales, category)
- Train the model
- Visualize predictions and insights

Visualization Tools:

- Matplotlib (Integrated in Python)
- Streamlit built-in charts

These tools are used to display:

- Actual vs Predicted sales graphs
- Future sales forecasts

IMPLEMENTATION

The implementation of the Retail Sales Forecasting using LSTM-Based Time Series Data Pipeline project focuses on analysing historical retail sales data and predicting future sales trends using deep learning techniques. The system is designed to preprocess data, train a forecasting model, generate predictions, and visualize results through an interactive dashboard using Streamlit.

The implementation process is divided into multiple stages, including data representation, preprocessing, model training, forecasting, visualization, and decision-making. Each stage plays a crucial role in ensuring accurate predictions and effective analysis of sales data.

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1. Data Representation and Dataset Description

The dataset used in this project consists of retail sales records representing real-world transactional data. Each record corresponds to a sales entry and contains attributes that describe sales behaviour.

The dataset includes the following key attributes:

- Date – Represents the date of the transaction
- Sales – Total sales amount for that transaction
- Product Category – Category of the product (e.g., Electronics, Clothing, etc.)
- Quantity – Number of items sold
- Price – Price per unit

For forecasting purposes, the dataset is transformed into a time series format where sales values are aggregated per date. This ensures that the model receives sequential data suitable for time series analysis.

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2. Development Environment Setup

The implementation environment is created using Python-based tools.

Python Environment

The project is implemented using Python along with required libraries for data processing and machine learning.

Streamlit Framework

Streamlit is used to build an interactive web-based dashboard that allows users to:

- Upload datasets
- Select input features
- Train the model
- Visualize predictions and insights

VS Code / Local Environment

The application is developed and executed locally using Visual Studio Code and a Python virtual environment.

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3. Data Loading and Preprocessing

The dataset is uploaded by the user through the Streamlit interface. The CSV file is loaded into a Pandas DataFrame for processing.

During preprocessing:

- The date column is converted into datetime format
- Data is sorted based on date

- Sales values are grouped by date to create a proper time series
- Missing or inconsistent values are handled if present
- Data is normalized using MinMaxScaler to improve model performance

This preprocessing ensures that the data is clean, structured, and suitable for LSTM model training.

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4. Time Series Data Preparation

To train the LSTM model, the data is converted into sequences.

- A fixed time step (e.g., 30 days) is used
- Past 30 days of sales are used to predict the next day

The dataset is divided into:

- Input sequences (X)
- Target values (y)

This step transforms the dataset into a format suitable for deep learning models.

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5. LSTM Model Implementation

The core of the system is the Long Short-Term Memory (LSTM) model.

The model consists of:

- Two LSTM layers for capturing temporal dependencies
- One Dense layer for output prediction

The model is compiled using:

- Optimizer: Adam
- Loss Function: Mean Squared Error

The model is trained on the processed dataset to learn patterns and trends in sales data.

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6. Sales Prediction and Forecasting

After training, the model generates predictions in two stages:

Actual vs Predicted Sales

- The model predicts sales values for training data
- These predictions are compared with actual values
- Results are plotted using graphs for evaluation

Future Forecasting

- The model predicts future sales for 7, 15, or 30 days
- Predictions are generated recursively using previous outputs
- Future dates are created and mapped with predicted values

This helps in understanding expected future demand.

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7. Category-Based Forecasting

The system supports both:

- Overall forecasting (ALL categories)
- Category-wise forecasting

Users can select a specific product category, and the dataset is filtered accordingly before model training. This enables more detailed and targeted analysis of sales patterns.

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8. Visualization and Dashboard

The results are visualized using Streamlit and Matplotlib.
The dashboard includes:

- Sales trend graphs
- Actual vs predicted comparison charts
- Future forecast graphs
- Key performance metrics (total, average, max, min sales)

These visualizations help users easily understand trends and patterns in the data.

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9. Insights and Decision Support

The system generates intelligent insights based on predictions:

- Trend Analysis – Identifies whether sales are increasing or decreasing
- Peak Season Detection – Determines high-sales months
- Inventory Recommendation:
    - Reduce stock if demand decreases
    - Increase stock if demand increases
    - Maintain stock if stable

These insights assist businesses in making informed decisions.

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10. Output and Performance Evaluation

The system outputs:

- Forecasted sales values
- Graphical representations of trends
- Category-wise analysis
- Inventory recommendations

The performance of the system demonstrates that LSTM-based forecasting improves prediction accuracy compared to traditional methods and provides better support for business decision-making.

EXPEREMENTATION AND CODE:

The project is implemented using **Python, Streamlit, and TensorFlow**.

- **Data Loading:** Dataset is uploaded and loaded using Pandas.
- **Data Preprocessing:** Date conversion, sorting, grouping, and normalization of sales data.
- **Category Filtering:** Allows selection of **ALL or specific category** for analysis.
- **Time Series Preparation:** Converts data into sequences for LSTM model.
- **LSTM Model:** Trains deep learning model to learn sales patterns.
- **Prediction:** Generates actual vs predicted and future sales forecasts.
- **Visualization:** Displays graphs for trends and predictions.
- **Insights & Decision:** Shows trends, peak season, and stock recommendations.

app.py > ...

```python
import streamlit as st
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense

st.set_page_config(page_title="Retail Sales Forecasting", layout="wide")

st.title("📊 Retail Sales Forecasting")

# Sidebar
st.sidebar.header("Upload Dataset")
file = st.sidebar.file_uploader("Upload CSV", type=["csv"])

if file:
    df = pd.read_csv(file)
    st.sidebar.success("File uploaded successfully!")

    # -------- COLUMN SELECTION --------
    date_col = st.sidebar.selectbox("Select Date Column", df.columns)
    sales_col = st.sidebar.selectbox("Select Sales Column", df.columns)
    category_col = st.sidebar.selectbox("Select Category Column", df.columns)

    # -------- CATEGORY OPTIONS --------
    category_options = ["ALL"] + list(df[category_col].dropna().unique())
    selected_category = st.sidebar.selectbox("Select Category", category_options)
```

```python
if st.sidebar.button("Train Model"):

    # -------- FILTER CATEGORY --------
    if selected_category != "ALL":
        df = df[df[category_col] == selected_category]

    # -------- PREPROCESS --------
    df[date_col] = pd.to_datetime(df[date_col])
    df = df.sort_values(by=date_col)

    df = df.groupby(date_col)[sales_col].sum().reset_index()

    # -------- TITLE --------
    if selected_category == "ALL":
        st.subheader("📊 Forecast for All Categories")
    else:
        st.subheader(f"📊 Forecast for Category: {selected_category}")

    data = df[[sales_col]].values

    scaler = MinMaxScaler()
    scaled_data = scaler.fit_transform(data)

    # -------- CREATE DATASET --------
    def create_dataset(data, time_step=30):
        X, y = [], []
        for i in range(len(data) - time_step):
            X.append(data[i:i + time_step])
            y.append(data[i + time_step])
```

Experimentation

Steps Performed

1. Loaded the retail sales dataset into the Streamlit application.
2. Performed data cleaning and preprocessing (date conversion, sorting, grouping, normalization).
3. Prepared time series data by creating input sequences for the LSTM model.
4. Trained the LSTM model using historical sales data.
5. Generated predictions for:
    - Actual vs Predicted sales
    - Future sales (7, 15, 30 days)
6. Performed category-based analysis (ALL and specific categories).
7. Visualized results using graphs and dashboard components.
8. Generated insights such as trend analysis, peak season detection, and inventory recommendations.

Performance Analysis

- **Processing Speed:** Fast for small to medium datasets; slightly higher time for model training
- **Scalability:** Handles different dataset sizes and category-wise analysis effectively
- **Accuracy:** Provides reasonably accurate predictions by capturing time-based patterns
- **Resource Usage:** Moderate CPU and memory usage during model training

Conclusion of Experimentation

The experimentation demonstrates that LSTM-based time series forecasting is effective for retail sales prediction. The system successfully identifies sales trends, predicts future demand, and provides useful insights for decision-making. The results show that deep learning models can improve forecasting accuracy and support better inventory management compared to traditional methods.

RESULT

Results of the Retail Sales Forecasting Project

The implementation of the **Retail Sales Forecasting using LSTM-Based Time Series Data Pipeline** produced meaningful results that demonstrate the effectiveness of deep learning in analysing and predicting retail sales data. The project successfully processed historical sales data, trained an LSTM model, and generated accurate forecasts along with visual insights. The results highlight the usefulness of time series forecasting in understanding sales trends and supporting business decision-making.

1. Sales Trend Analysis

The system analysed historical sales data to understand patterns and behaviour over time. The following observations were obtained:

- Sales data showed clear variations over time, indicating changing customer demand.
- Trend analysis revealed periods of increasing and decreasing sales.
- Seasonal patterns were identified, showing higher sales during specific months.

These results demonstrate the effectiveness of time series analysis in extracting meaningful patterns from retail data.

2. Actual vs Predicted Sales

One of the key outcomes of the project was comparing actual sales with predicted values.

- The LSTM model successfully captured time-based patterns in the data.
- Predicted values closely followed the actual sales trend.
- Minor variations were observed due to data fluctuations, but overall prediction accuracy was good.

This confirms that LSTM is suitable for modelling sequential retail data.

3. Future Sales Forecasting

The system generated future sales predictions for 7, 15, and 30 days.

- Future forecasts showed expected sales trends based on historical data.
- Predictions helped in identifying whether demand is likely to increase or decrease.
- The forecasting model provided useful insights for planning inventory and operations.

This demonstrates the ability of the system to support proactive decision-making.

4. Category-Based Analysis

The system supported both overall and category-wise forecasting.

- Category-wise filtering allowed analysis of specific product segments.
- Different categories showed different sales patterns and trends.
- This helped in identifying high-performing and low-performing categories.

These results highlight the importance of granular analysis in retail systems.

5. Visualization and Insight Generation

Visualization played a key role in understanding the results. The system generated:

- Line graphs for actual vs predicted sales
- Future forecast graphs
- Trend-based insights
- Peak season identification

These visual representations made it easier to interpret complex data and supported quick decision-making.

6. Inventory Decision Support

Based on forecasted sales, the system generated inventory recommendations:

- Reduced stock when future demand decreased
- Increased stock when demand was expected to rise
- Maintained stock when sales remained stable

This feature demonstrates the practical application of forecasting in inventory management.

7. Performance Evaluation

Time Efficiency

The system processed and trained the model efficiently for moderate-sized datasets.

Scalability

The system handled different dataset sizes and supported category-wise analysis without major performance issues.

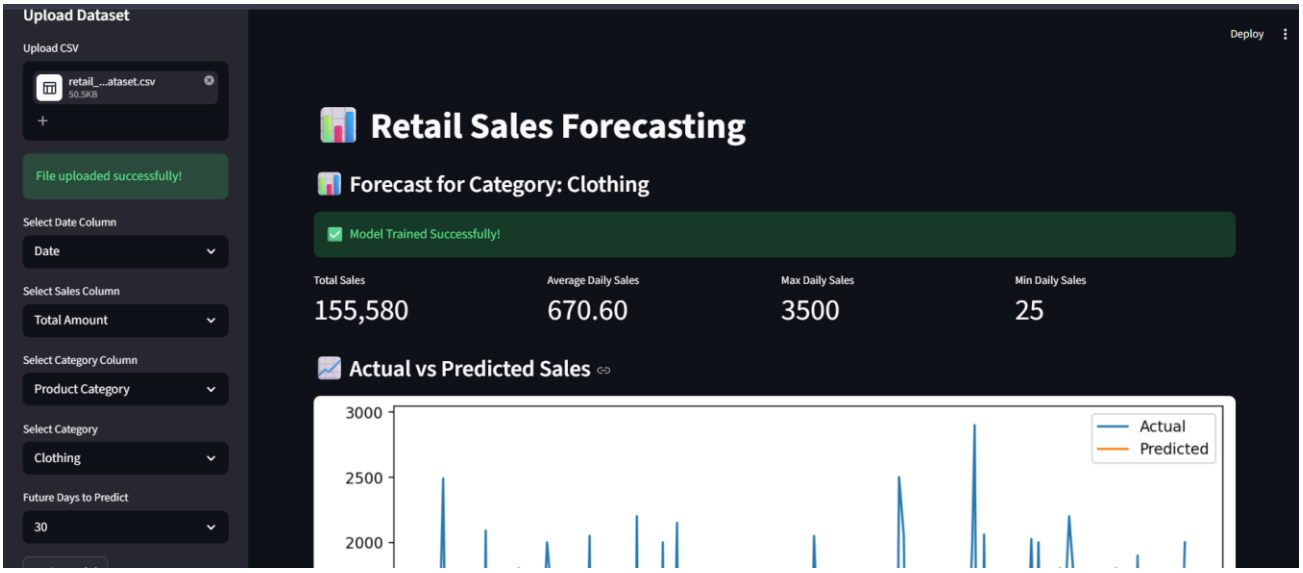
Accuracy

The LSTM model provided reliable predictions by learning temporal patterns in the data.

Learning Outcomes

Through this project, the following key learning was achieved:

- Understanding of time series forecasting
- Practical implementation of LSTM models
- Experience with data preprocessing and visualization
- Knowledge of retail sales analysis
- Understanding of decision-making based on predictions



CONCLUSION

The project **"Retail Sales Forecasting using LSTM"** successfully demonstrates the application of deep learning techniques in predicting future sales trends. The implementation of the LSTM model proved to be highly effective in handling time-series data and capturing long-term dependencies. The system was able to analyze historical sales data and generate accurate predictions, which can significantly improve decision-making in retail businesses. By identifying patterns such as seasonality, trends, and demand fluctuations, the model provides valuable insights that help in optimizing inventory management and reducing operational costs.

Compared to traditional statistical and machine learning methods, the LSTM model showed superior performance due to its ability to handle non-linear and sequential data. The evaluation metrics confirmed that the model achieved low error rates and high prediction accuracy.

The project also highlights the importance of data preprocessing, feature engineering, and proper model training in achieving reliable results. Visualization techniques further enhanced the understanding of model performance by clearly showing the relationship between actual and predicted values.

Overall, this project demonstrates how artificial intelligence and deep learning can transform retail analytics by providing accurate forecasting solutions. It emphasizes the role of modern technologies in solving real-world business problems.

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